



JURONG JUNIOR COLLEGE
2017 JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE
NAME

CLASS

17S

EXAM INDEX

CHEMISTRY

9729/02

Paper 2 Structured Questions

29 August 2017

2 hours

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name, class and exam index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a HB pencil for any diagrams, graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

A *Data Booklet* is provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	10
2	18
3	11
4	10
5	12
6	14
Penalty (delete accordingly)	
Lack 3sf in final ans	-1 / NA
Missing/wrong units in final ans	-1 / NA
Bond linkages	-1 / NA
Total	

This document consists of **17** printed pages and **1** blank page.

Answer **all** the questions.

For
Examiner's
Use

- 1 (a) Draw a fully labelled diagram of the electrochemical cell you would set up in order to measure the standard electrode potential of the Cu^{2+}/Cu half-cell.

[3]

In the Earth's crust, copper often occurs as the sulfide associated with zinc sulfide. Pure copper is extracted from this mixture.

In the first stage of extraction, the copper extracted contains some zinc and silver impurities. Electrolysis was used to purify the extracted copper with impure copper as the anode, and pure copper at the cathode.

- (b) Using relevant data from the *Data Booklet*, explain what happens to the impurities at the anode.

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[2]

- 1 (c) Using relevant data from the *Data Booklet*, explain what happens at the cathode.

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[2]

- (d) A similar set up can be used for copper plating, using pure copper electrode as the anode.

Calculate the time taken to coat 0.50 g of copper on the object if a current of 0.20 A was passed through the electrolysis bath.

[3]

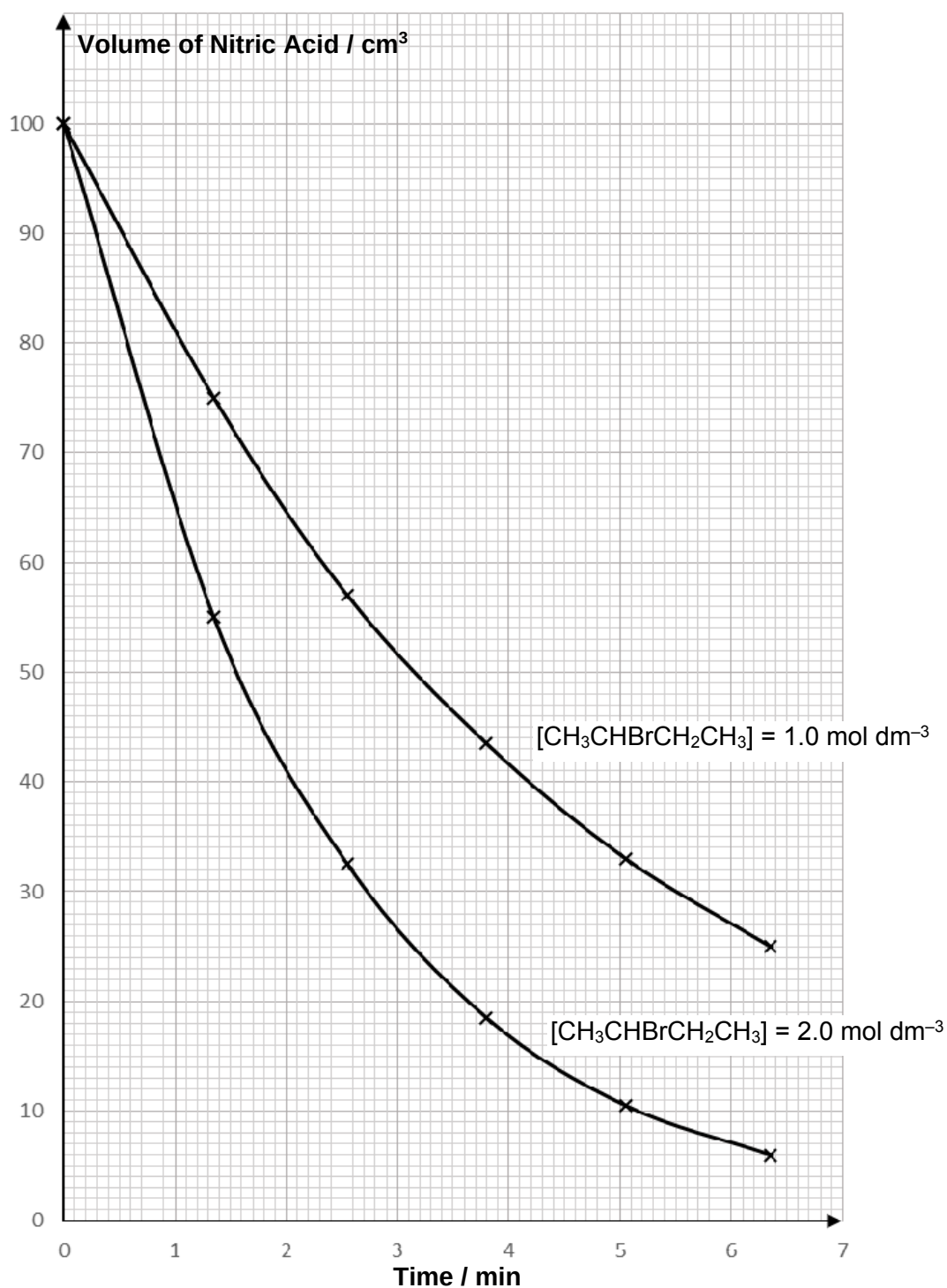
[Total: 10]

- 2 (a) 2-bromobutane, $\text{CH}_3\text{CHBrCH}_2\text{CH}_3$, is hydrolysed by aqueous sodium hydroxide.

The kinetics of the reaction can be monitored by withdrawing samples of the solution at appropriate times and titrating them with aqueous nitric acid after quenching.

Two sets of experiments were performed using different initial concentrations of the 2-bromobutane while the initial concentration of sodium hydroxide was kept at 0.1 mol dm^{-3} .

The results obtained are shown in the graphs below.



- 2 (a) (i) Using the graph where $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3] = 1.0 \text{ mol dm}^{-3}$, find the order of reaction with respect to aqueous sodium hydroxide.

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[2]

- (ii) Using the graphs above, deduce the order of reaction with respect to 2-bromobutane.

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[3]

- (iii) Hence, write down the rate equation for this reaction.

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[1]

- (iv) State the half-life of sodium hydroxide when $[\text{NaOH}] = 0.2 \text{ mol dm}^{-3}$ and $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3] = 1.0 \text{ mol dm}^{-3}$.

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[1]

- (v) Calculate the time taken for $[\text{NaOH}]$ to drop to 1 % of its original value when $[\text{CH}_3\text{CHBrCH}_2\text{CH}_3] = 1.0 \text{ mol dm}^{-3}$.

[1]

- 2 (b) (i) With reference to (a)(iii), name and describe the mechanism of this hydrolysis reaction.

Type of mechanism:

[3]

- (ii) Draw the energy profile diagram for the mechanism you have described in (b)(i).

On the diagram, label

- the reactants and products
- the enthalpy change of reaction, ΔH ,
- the activation energy, E_a



[3]

- (iii) How would the rate change if the experiment is repeated using 2-chlorobutane instead of 2-bromobutane? Explain.

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[2]

- 2 (b) (iv) State and explain if the resultant mixture will be optically active if the reaction started with an optically pure sample of 2-bromobutane.

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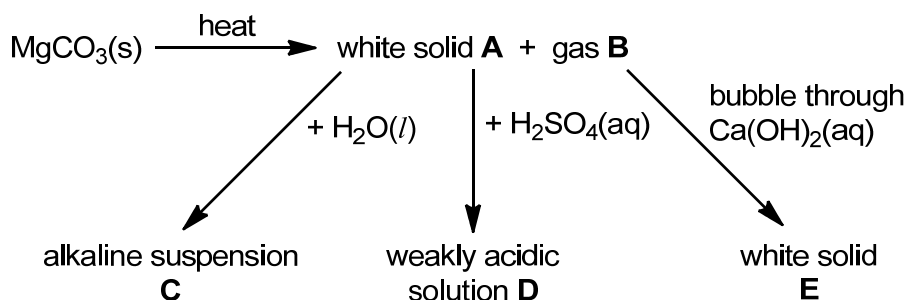
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[2]

[Total: 18]

- 3 Group 2 elements are ingredients in many important materials from cement to face cream.

A student carried out the following experiments as shown in the reaction scheme below using a sample of magnesium carbonate.



- (a) Identify compounds A to E in the reaction provided.

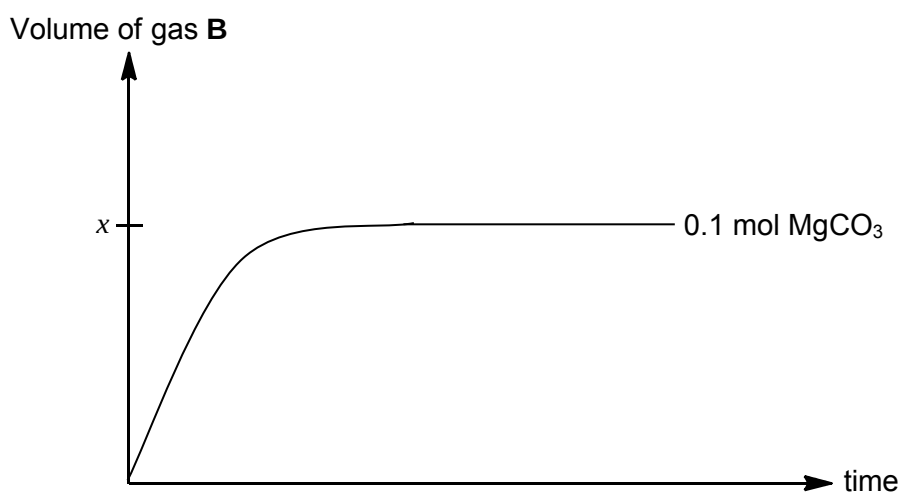
A: B: C: D: E: [5]

- (b) Explain, with the aid of an equation, why solution D is weakly acidic.

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 [2]

- (c) The graph below shows the volume of gas B collected over time when 0.1 mol of magnesium carbonate is heated.



- (i) Draw, on the same axes above, the graph that would be obtained when 0.1 mol barium carbonate was used and heated until it decomposed completely.

[1]

3 (c) (ii) Explain the shape of the graph that you have drawn for (c)(i).

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[3]

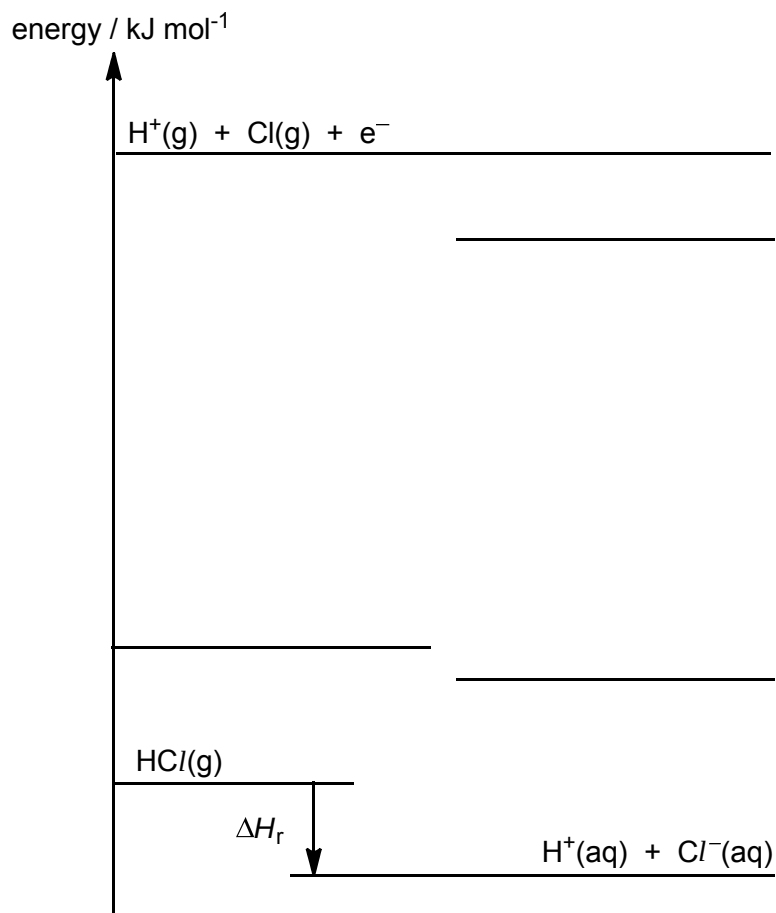
[Total: 11]

- 4 (a) Using relevant data from the *Data Booklet* and the following data, construct a labelled energy level diagram to calculate the enthalpy change of solution for gaseous hydrogen chloride.

Electron affinity of chlorine = -364 kJ mol^{-1}

Enthalpy change of hydration of hydrogen ion = $-1071 \text{ kJ mol}^{-1}$

Enthalpy change of hydration of chloride ion = -381 kJ mol^{-1}



[3]

- (b) Hence, calculate ΔG_{soln} for hydrogen chloride, given that $T\Delta S_{\text{soln}}$ is -18 kJ mol^{-1} .

[1]

- 4 (c) (i) Draw a labelled diagram to illustrate the interaction between chloride ion and water. State the type of interaction clearly in your diagram.

[2]

- (ii) Explain how the magnitude of enthalpy change of hydration of bromide ions differs from that of chloride ions.

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[2]

- (e) *Use of Data Booklet is relevant to this question.*

Both hydrogen chloride and hydrogen bromide dissolve readily in water.

Describe a chemical test, which does not involve silver nitrate, that could distinguish between aqueous solutions of these two gases.

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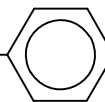
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[2]

[Total: 10]

- 5 Amino acids are critical to life, and they serve as the building blocks of proteins. Out of the vast variety of amino acids, twenty are termed as essential because they cannot be created by the human body, and therefore must be supplied in the diet.

A peptide **A** was analysed and contained the following amino acids, $[RCH(NH_2)CO_2H]$.

amino acid	abbreviation	formula of R group	number of residues
aspartic acid	asp	$-CH_2COOH$	one
glycine	gly	$-H$	one
serine	ser	$-CH_2OH$	two
tyrosine	tyr	$-CH_2-$  $-OH$	one
valine	val	$-CH(CH_3)_2$	one

- (a) Suggest suitable reagent and conditions for complete hydrolysis of proteins to amino acids non-enzymatically.

..... [1]

- (b) (i) Given that aspartic acid has pK_a values of 2.0, 3.9 and 9.9, draw the structural formula of aspartic acid in its most stable state at pH 7.

[2]

- (ii) Electrophoresis was used to separate the amino acids in the mixture. Suggest, if any, whether aspartic acid will migrate towards the cathode or the anode.

..... [1]

- (c) (i) Draw a structure of the dipeptide with the sequence, ser-asp.

[2]

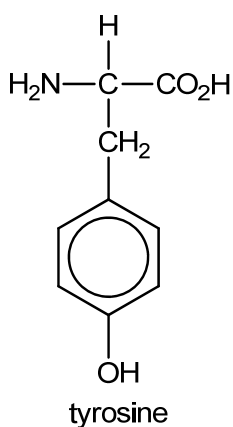
5 (c) (ii) Analysis of peptide **A** gave the following results:

- The N-terminus was shown to be ser.
- On reaction with a special reagent which digests at the carboxylic end of val, two peptides were obtained. One of these two was a dipeptide of sequence gly-ser.
- The enzyme chymotrypsin, which hydrolyses at the carboxylic group of tyr, gave two tripeptides.

Use these results to deduce the amino acid sequence of peptide **A**.

..... [2]

(d) Give the organic product(s) formed when tyrosine is subjected to the following reagents and conditions separately.



Reagent and Conditions	Organic Product(s)
Aqueous Br ₂	
Cold aqueous NaOH	
CH ₃ COOH	

[4]

[Total: 12]

6 This question is about the reactions of Group 17 elements.

- (a) 10.0 cm³ of an aqueous solution containing 0.10 mol dm⁻³ Br₂ was added to 50.00 cm³ of a 0.10 mol dm⁻³ NaOH solution.

The resulting solution required 30.00 cm³ of 0.10 mol dm⁻³ HCl for complete neutralisation.

- (i) Calculate the mole ratio of Br₂ to NaOH.

[3]

- (ii) It was observed that during the reaction between Br₂ and NaOH, there is no gas evolved.

Given that the products formed are Br⁻ and BrO_x⁻, write a balanced ionic equation for the reaction and hence deduce the value of *x*.

Balanced ionic equation:

Value of *x*:

[1]

- 6 (b) Mass spectrometry is an analytical technique which measures the masses of chemical species present within a sample. A mass spectrum can be used to determine the isotopic composition of a sample, masses of chemical species and to elucidate the chemical structure of a chemical compound.

The mass-to-charge ratio, also known as m/e value, is the M_r of chemical species present in the sample. The peak height ratio gives an indication of the isotopic ratio of elements in the sample.

In two separate experiments, propene was reacted with bromine in inert organic solvent and with aqueous bromine. The organic products were then extracted and analysed using mass spectrometry. The graphs below show the mass spectrum for each experiment.

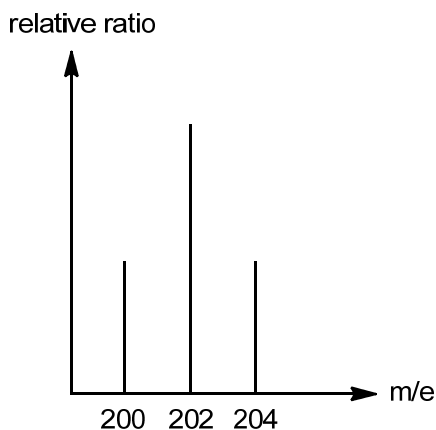


Fig 2.1: reaction with Br_2 in inert solvent

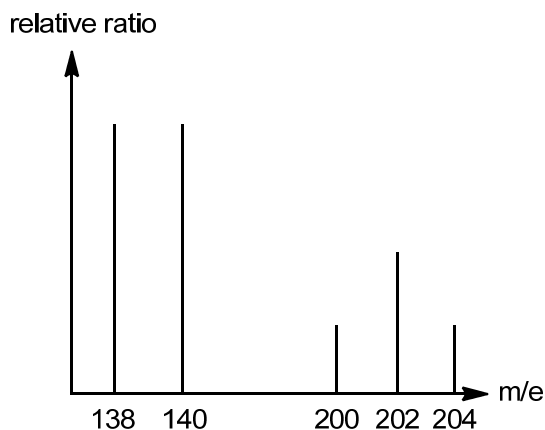


Fig 2.2: reaction with $\text{Br}_2(\text{aq})$

- (i) Draw the structure of the product formed when propene reacts with bromine in inert organic solvent.

[1]

- (ii) Using your answer in (b)(i) and Figure 2.1, determine the isotopic composition of bromine.

You may assume that carbon, hydrogen and oxygen only exist in one isotopic form of ^{12}C , ^1H and ^{16}O respectively.

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[1]

- 6 (b) (iii) Name and outline the mechanism for the reaction between propene and bromine in inert organic solvent.

Name of mechanism:

[4]

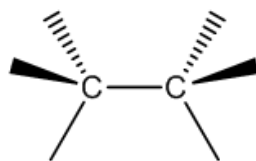
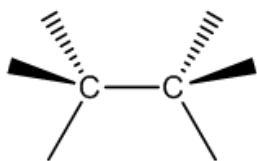
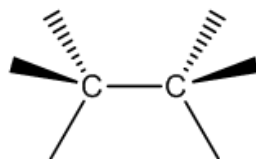
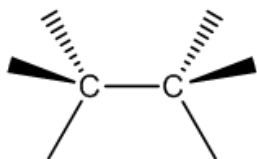
- (c) When propene is reacted with aqueous bromine, a different mass spectrum was obtained as shown in Fig 2.2. There are two additional peaks with greater peak height at m/e values of 138 and 140 than those at m/e values of 200 and above.

- (i) Explain, with reference to the mechanism you have described in (b)(iii), why the peaks at m/e values of 138 and 140 have greater peak height.

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[2]

- 6 (c) (ii) Draw the four different possible products that are responsible for the peak at m/e value of 138 by completing the diagram below.



[2]

[Total: 14]

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